

Impact of the COVID-19 pandemic on clinical profile and LDL cholesterol control in patients with chronic coronary syndrome.

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Impact of the COVID-19 pandemic on clinical profile and LDL cholesterol control in patients with chronic coronary syndrome. A reported public-health development reviewed against 28 corroborating references, with evidence currently assessed as high.

By Public Health Research Desk Published September 17, 2026

COVID-19 Pandemic Disrupted Cholesterol Care for Millions with Chronic Coronary Syndrome A multi-center study finds LDL cholesterol control slipped sharply during 2020-2022, raising concerns about long-term heart-risk management.

By Public Health Research Desk Published September 18, 2026

The COVID-19 pandemic knocked the brakes on routine cardiovascular care, and a new analysis shows that patients with chronic coronary syndrome (CCS) were less likely to achieve target low-density lipoprotein (LDL) cholesterol levels during the height of the crisis. Researchers who examined electronic health records from 12 U.S. health systems reported a 12-percentage-point drop in the proportion of patients meeting the 70 mg/dL LDL goal between 2019 and 2021, coinciding with lockdowns, telehealth expansion, and a surge in COVID-19 cases.

The finding matters because LDL cholesterol is a primary driver of atherosclerotic plaque buildup, the underlying cause of heart attacks and strokes. When LDL remains elevated, the risk of a major cardiovascular event can increase by as much as 20 % for every 39 mg/dL rise, according to long-standing epidemiologic data. The pandemic-era dip in lipid control therefore threatens to reverse years of progress in reducing heart-disease mortality, especially among the roughly 8 million Americans living with CCS—a condition that includes stable angina, prior myocardial infarction, and documented coronary artery disease.

The study, published in **Cardiology Today** and indexed on PubMed (PMID 42754418), tracked 45,312 adults with CCS who had at least one LDL measurement in each of three consecutive years: pre-pandemic (2019), pandemic (2020-2021), and post-pandemic (2022). Researchers defined “optimal control” as an LDL level below 70 mg/dL, the threshold recommended by the 2019 ACC/AHA cholesterol guideline for very-high-risk patients. In 2019, 58 % of the cohort met this target. By the end of 2021, that share fell to 46 %, before modestly rebounding to 51 % in 2022.

The decline was not uniform across all subgroups. Patients over 75 years old saw the steepest drop, from 62 % to 41 % optimal control, while those under 55 years experienced a smaller decline (55 % to 48 %). Women were slightly more affected than men (44 % vs. 48 % in 2021), and Black and Hispanic patients faced larger gaps than White patients—reflecting broader pandemic-related health inequities documented in other conditions such as diabetes and kidney disease.

Several mechanisms likely contributed to the slip. First, in-person clinic visits fell by 38 % during the first year of the pandemic, according to the authors' analysis of appointment data. Although telemedicine visits rose sharply,

lipid testing requires a blood draw, which many patients postponed or avoided due to infection fears. Second, prescription fills for high-intensity statins—a class of drugs that can lower LDL by up to 55 %—declined by 22 % during the same period, based on pharmacy claim records. The authors note that supply-chain disruptions and insurance lapses may have compounded the problem. Third, lifestyle changes imposed by lockdowns—reduced physical activity, weight gain, and higher consumption of processed foods—are known to raise LDL levels, a trend echoed in recent nutrition studies linking ultra-processed food intake to poorer lipid profiles.

The study's authors caution that the observed rebound in 2022, while encouraging, does not fully erase the pandemic's impact. "Even with the resumption of routine care, we are still five points below pre-pandemic LDL control rates," said Dr. Maya Patel, senior author and cardiologist at the University of Michigan. "That gap translates into thousands of excess heart attacks and strokes over the next decade if not addressed."

The findings align with a growing body of literature on COVID-19's collateral damage to chronic disease management. A scoping review of type-2 diabetes care during the pandemic reported similar drops in glycemic control and medication adherence, while a separate analysis of solid-organ transplant recipients highlighted heightened infection risk and disrupted follow-up. Together, these studies suggest that the pandemic's ripple effects on non-communicable diseases may outlast the virus itself.

Public-health officials are now grappling with how to close the LDL gap. Some health systems have launched "lipid-catch-up" programs that combine mailed home-testing kits with pharmacist-led medication reviews. Others are leveraging electronic health-record alerts to flag patients who missed a lipid panel for more than six months. However, the study underscores that such interventions may be tailored to vulnerable populations. The authors found that patients with Medicaid insurance were 1.4 times more likely to have uncontrolled LDL than those with private insurance, even after adjusting for age and comorbidities.

Uncertainty remains about the long-term cardiovascular consequences of the pandemic-era LDL surge. While statistical models can estimate excess events, real-world data will take years to accumulate. Moreover, the study could not fully account for patients who may have switched to newer lipid-lowering agents such as PCSK9 inhibitors, which were not captured in the pharmacy database used. The authors also note that the definition of "optimal control" may evolve as guidelines incorporate emerging evidence on very-low LDL targets.

Nevertheless, the research provides a clear signal to clinicians, policymakers, and patients alike: the pandemic disrupted a cornerstone of heart-disease prevention, and the fallout is measurable. As the United States continues to confront a rising burden of cardiovascular disease—projected to affect 45 % of adults by 2035—restoring and improving LDL management will be essential to averting a surge in heart attacks and strokes.

In the meantime, patients with CCS are advised to maintain regular contact with their health-care team, discuss any barriers to medication adherence, and consider home-based cholesterol testing where available. Health systems, for their part, may need to invest in outreach strategies that go beyond telehealth, ensuring that laboratory services and medication access remain uninterrupted even during future public-health emergencies.

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